

0.02 mm	7
SCAN ACCURACY	CORE SERVICES

6

INDUSTRIES SERVED

6+

COUNTRIES WITH REPEAT CLIENTS

Use one service on its own, or hand us the whole job — from first scan to final file.



03

Scan to BIM

Existing buildings captured as accurate, intelligent BIM models



04

Class A Surfacing

Flawless, curvature-continuous surfaces for visible parts

intelligent BIM models.

for visible parts.



07

Product Rendering

Photoreal visuals that sell a product before it is ever made.

End-to-end delivery

Scan, engineer, inspect, print and render — under one roof, with one point of contact.









03

Engineer

CAD, Class A surfaces, BIM or inspection built from that data.

04

Deliver

Native files, drawings and reports in the formats your team uses.

2.1) — SERVICES

3D Scanning

pture any part, product or space as an accurate digital model.

scan parts of every size with a range of professional 3D scanners — from all precision components to car bodies and full buildings. The result is a precise tal copy of the physical object, ready for product design, reverse engineering quality control.

h-accuracy capture

-laser and structured-light scanners record il down to hundredths of a millimetre.

Any size, on site or in-house

From jewellery to vehicles and plant rooms — we bring the scanner to you when the part can't

travel.

an, usable data
ns are aligned, cleaned and merged into
ertight meshes, not raw point dumps.

Ready for the next step
Data is prepared for reverse engineering, inspection, printing or visualisation.

IVERABLES

oint cloud (.PLY, .TXT) | Mesh (.STL, .OBJ) | Colour/texture scans | Scan report

USTRIES

erospace | Automobile | Healthcare | Jewellery | Education

2.2) — SERVICES

Reverse Engineering

rn scan data into fully editable, manufacturable CAD.

rebuild physical parts as parametric CAD models — for re-manufacturing
acy parts with no drawings, modifying existing designs, or creating moulds and
ing. You choose how faithful the model should be to the scanned part.

sign-intent modelling
ean CAD model with manufacturing
erfections removed — ideal for re-manufacture
n 3D models or 2D drawings.

NURBS surface models
Precise freeform surfaces fitted over scan data where an exact as-is representation matters.

orid modelling
matic features modelled to design intent, form areas surfaced to the scan — the best of .

Drawings on request
2D manufacturing drawings with GD&T generated from the final model.

IVERABLES

rametric CAD (.STEP, .IGES, .X_T) | Native SolidWorks / Creo / NX | 2D drawings (.PDF, .DWG) | eviation report vs. scan

USTRIES

tomobile | Aerospace | Healthcare

2.3) — SERVICES

Scan to BIM

isting buildings captured as accurate, intelligent BIM models.

laser-scan existing structures and convert the point cloud into detailed BIM dels. Architects, engineers and contractors get an accurate as-built foundation renovation, retrofit and facility-management projects — with every wall, slab, umn and service documented.

built accuracy
estrial laser scanning records the true metry of the building, not the original

LOD 200–400 models
Walls, floors, structure and MEP modelled to the level of detail your project needs.

vings.

sh-ready coordination
lels are set up for coordination with new gn work in Revit and Navisworks.

Drawings from the model
Floor plans, sections and elevations exported straight from the BIM.

IVERABLES

- gistered point cloud (.E57, .RCP)
- Revit model (.RVT)
- IFC export
- plans, sections, elevations

USTRIES

- onstruction
- Education
- Healthcare

· VSHARKSOLUTIONS.COM 07

2.4) — SERVICES

Class A Surfacing

wless, curvature-continuous surfaces for visible parts.

ss A surfaces are the visible skins of a product — car body panels, aircraft ngs, consumer-product housings. We build G2/G3-continuous surfaces with an patch layouts and verify them with zebra and curvature analysis, so ections flow without a ripple.

/ G3 continuity
vature-continuous transitions between every h for perfect reflections.

Analysis-verified
Zebra stripes, highlight lines and curvature combs checked on every surface.

m scan or sketch

t from a clay-model scan, a reference surface oncept sketches.

Tooling-ready

Surfaces handed over with draft, flanges and offsets ready for engineering.

IVERABLES

ass A surfaces (.IGES, .STEP)

Native Alias / ICEM models

Surface-analysis reports

nder-ready model

USTRIES

tomobile

Aerospace

2.5) — SERVICES

Digital Metrology

mpare manufactured parts to their CAD — to the micron.

inspect manufactured parts by scanning them and comparing the result inst the nominal CAD. Colour deviation maps, GD&T checks and first-article orts show exactly where a part is in or out of tolerance — fast, and without tact.

our deviation maps

surface comparison against CAD, instantly wing where material is over or under.

GD&T inspection

Flatness, position, profile and more, measured and reported against the drawing.

st-article inspection
mplete FAI reports for new parts, tools and pliers.

Trend analysis
Repeated inspections track tool wear and process drift over time.

IVERABLES

spection report (.PDF) | Colour deviation maps | GD&T results tables | Scan data for records

USTRIES

erospace | Automobile | Healthcare

2.6) — SERVICES

SD Printing

pid prototypes and functional parts, straight from your model.

print prototypes, fit-check models, jigs, fixtures and short-run end-use parts ctly from CAD or scan data. Our team also prepares and repairs files — uding sculpted character models — so they print right the first time.

pid prototyping
sical parts in days to check form, fit and tion before tooling.

Functional materials
Engineering plastics and resins chosen for strength, heat or detail.

nt-ready preparation
h repair, hollowing, supports and orientation dled for you.

Character & figure models
Digitally sculpted busts and figures prepared for high-detail resin printing.

IVERABLES

inted parts | Print-ready files (.STL, .3MF) | Finishing / painting on request | nall production batches

USTRIES

ucation | Healthcare | Jewellery | Automobile

2.7) — SERVICES

Product Rendering

otoreal visuals that sell a product before it is ever made.

turn CAD and design models into photorealistic images and animations for rketing, presentations and design reviews. Materials, lighting and scenes are ed so your product looks real — and looks its best — before the first unit is t.

otoreal stills
lio and lifestyle shots with accurate materials, hes and lighting.

Animations & turntables
Exploded views, assembly sequences and 360° spins.

eriors & exhibitions
ces, stands and environments visualised
re construction.

Jewellery visualisation
Metals and gemstones rendered with true reflections and sparkle.

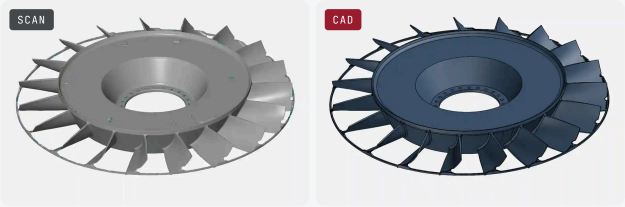
IVERABLES

gh-res stills (.PNG, .TIFF) | Animations (.MP4) | 360° turntables | Web-ready image sets

USTRIES

wellery | Construction | Automobile

Real parts we have captured and rebuilt as editable, manufacturable CAD.

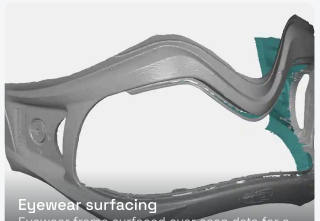


Impeller Bladed impeller captured and modelled with precise freeform blade surfaces.



Bumper Vehicle bumper scanned and resurfaced for tooling.

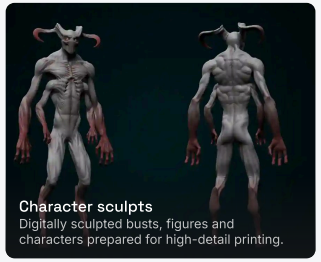
Surfacing, sculpting and photoreal rendering — visuals that sell a product before it is made.



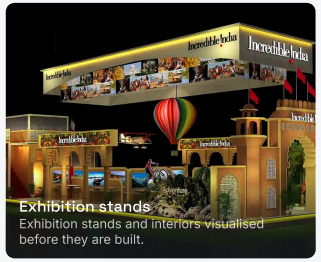


Eyewear frame surfaced over scan data for a custom-fit product.

Rings rendered with true metal reflections and gemstone sparkle.



Character sculpts
Digitally sculpted busts, figures and characters prepared for high-detail printing.



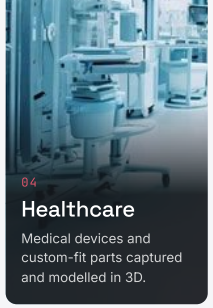
Exhibition stands
Exhibition stands and interiors visualised before they are built.

Six sectors, one standard of precision.





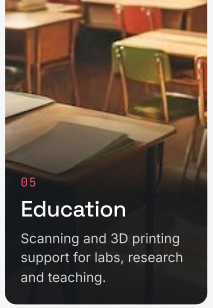




04

Healthcare

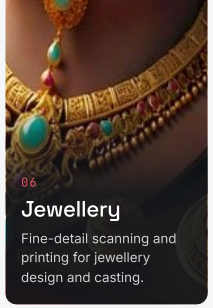
Medical devices and custom-fit parts captured and modelled in 3D.



05

Education

Scanning and 3D printing support for labs, research and teaching.



06

Jewellery

Fine-detail scanning and printing for jewellery design and casting.

Vertex blue-laser and optical-tracking scanners — now available from VShark, with training and support.

ex3D Pro

ld blue-laser 3D scanner

20 mm precision



Vertex Tracker

Optical-tracking 3D scanning system

0.025 mm precision

VERTEX3D PRO	VERTEX TRACKER

50 crossed blue laser lines	50 crossed blue laser lines
7,100,000 measurements/s	7,100,000 measurements/s
650 × 550 mm	1100 × 800 mm
0.01 mm	0.01 mm
0.015 mm + 0.035 mm/m	10.0 m^3: 0.060 mm · 25.0 m^3: 0.083 mm
> 10 mm	10 mm – 8000 mm
.STL, .PLY, .ASC, .TXT	.STL, .PLY, .ASC, .TXT

GB+ VRAM · Intel Core i7-13700H or better · 32 GB RAM or more

US · UK · GERMANY · CANADA · UAE · SAUDI ARAB

engineering service
heir attention to detail and
e working with them a
ed with them many times

“

Working with VSHARK on a reverse engineering and 3D modelling project was impressive; their professionalism, clear communication, and dedication to quality made the entire process smooth and efficient.

Benedikt Klein

Benedikt Klein
GERMANY

exceptional work beyond
ention to detail and
entire process seamless. I
r any project.

“

Working with VSHARK Engineers was a fantastic experience. Their creative approach and attention to detail perfectly aligned with our brand, and they were always polite and cooperative throughout the project.

Mariam Al-Ghayousi
DUBAI

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